

An Overview of Training for Climbing

It's the journey, experience, and self-discovery in climbing that hold the greatest value. But climbing hard routes is a heck of a lot of fun, too!

Many words can describe the wonderful activity of rock climbing—elegant, powerful, rewarding, and, sometimes, frustrating. While there may be nothing more natural and intuitive than climbing (just watch how children climb around on everything in sight!), rock climbing is indeed a complex activity with demands unique from those of living and playing in the everyday, horizontal world.

Performing in the vertical plane requires physical capabilities such as strength, power, flexibility, and endurance. It also demands the development of technical skills such as balance and economic movement while gripping and stepping in an infinite variety of ways, positions, and angles. Most important, the inherent stress of climbing away from the safety of the ground requires acute control of your thoughts, focus, anxiety, and fears. In aggregate, the above factors dovetail into what may be one of the more complex sporting activities on this third rock from the sun.

The goal of this book is to explore all the topics relevant to increasing the effectiveness of your training and the quality of your climbing. As a climber of forty years (who's been fortunate enough to meet and climb with many brilliant individuals), I feel the journey should begin with a primer on the history of training for rock climbing. Clearly, the advancements we make today are possible

only because we are standing on the shoulders of the giants who preceded us. Next we'll explore the interesting subject of genetics and the possible genetic limitations to climbing performance. This will lead us into an overview of training for climbing and the things you should consider in your quest for the biggest gains in performance in the shortest possible time.

Training for Climbing: A Brief History

Compared with many other sports, the science of performance rock climbing is still quite young. Well over a hundred years of literature exists on technical aspects of the golf swing, and Olympic sports have been the subject of performance analysis for centuries. Far removed from the mainstream of organized sports, and an almost countercultural pursuit just a generation ago, rock climbing was completely off the map in the emergence of sports sciences. What little information did exist on the technical aspects of climbing was mainly passed on by word of mouth in the form of tips on technique and equipment.

Nevertheless, some climbers used basic gymnastics, weight training, elementary bouldering, and building to either emulate actual climbing moves or gain the strength to perform at higher levels of the sport. Oscar Eckenstein, a Brit of Teutonic heritage and possibly the first documented boulderer, climbed ropes in the gym, did one-arm pull-ups, and pushed himself on small rocks during



John Gill's amazing one-arm front lever. JOHN GILL COLLECTION

the 1890s; George Leigh Mallory was a high-bar enthusiast and one of the first to do giant swings; E. A. Baker in *Moors, Caves, and Crags* (1903) tells of a colleague who “ascends the outside of an iron staircase on his fingers . . . and crosses in a sitting posture the tie-bars of a lofty roof”; Claude E. Benson in *British Mountaineering* (1909) talks of being “blessed with a basement staircase of stone . . . I am to be found hanging by my fingertips to the outside thereof.” And a gymnastic exercise of the nineteenth century involved climbing the underside of an oblique ladder using arms only—a precursor of the Bachar Ladder and campus training.

Given the extreme and run-out technical climbs being done on the Elbe River sandstone near Dresden a century ago—the hardest of which are now recognized as being near 5.10 difficulty—it is reasonable to conclude that early German free climbers placed a high value on style and difficulty. It is hard to conceive of such sustained routes being

climbed without some specific regime to build forearm and upper-body strength, although working routes on toprope may have been their primary method of training.

The strongest climbers of the early and mid-1900s included Oliver Perry-Smith, Albert Ellingwood, Joe and Paul Stettner, Fritz Wiessner, Jack Durrance, Hans Kraus, John Salathe, and Harold Goodro, as well as some of the early Yosemite masters such as Warren Harding, Dave Rearick, Bob Kamps, and Royal Robbins. All were natural athletes or had competed in other athletic activities prior to becoming climbers. More important, they all possessed a great sense of adventure and daring—a hallmark trait of all great climbers of this era. Mike Sherrick, Robbins's companion on the first ascent of the Northwest Face of Half Dome, was an excellent gymnast who often backflipped to the ground after finishing a boulder problem, much to the chagrin of his tamer companions. Yet training as a rock

climbing discipline built on vision, specificity, and intention was the innovation of a young man from Alabama who began climbing in the early 1950s.

Now one of the undisputed legends of American climbing, John Gill is the first person known to engage in highly regimented training for climbing. Unlike the others of his day who pushed themselves on vertical crags and long rock routes in the mountains, Gill—although an alpinist and rock climber—spent more of his time on short, overhanging faces on low boulders at the base of mountains or in river valleys. Bagging summits and climbing big walls had less aesthetic appeal for Gill; he instead sought the kinesthetics of dynamic movement up overhanging rock and adroitly built a novel training program to suit.

For more than fifteen years beginning in the mid-1950s, Gill trained on a gym rope, the still rings, and with weighted, fingertip pull-ups, one-arm and one-finger pull-ups, and one-arm front levers, in preparation for his powerful bouldering ascents throughout the Midwest, Southeast, and Rocky Mountains. In the early years Gill's gymnastic moves and the extremely muscular problems they produced—even his use of gymnasts' chalk—were viewed by most climbers with bemusement, if not bewilderment. Today his legacy as an innovator, visionary, and, in fact, the father of both modern bouldering and training for climbing is the foundation that has allowed route ratings to move into 5.13 and beyond. Gill's technical ability was years ahead of everyone else, as illustrated by his very bold 1961 free-solo first ascent of The Thimble in South Dakota's Black Hills, an overhanging 30-foot inspiration now rated V4 (5.12a), and his improbable center problem (incredibly, grade V9 by modern standards!) up the Red Cross Boulder in the Tetons two years before The Thimble. Unrecognized and underappreciated at the time, Gill in establishing these standards was an early prototype of today's top-end rock gymnasts, characterized like them by precise footwork, intense focus, and awesome power.

By the mid-1960s a number of other climbers, most with a background in gymnastics, also began training specific to climbing. Pat Ament, a young gymnast from Colorado, was an early training enthusiast and went on to become a leading climber and prolific developer of hard boulder problems. In 1967 Ament and Gill began a long friendship, and these two powerful boulderers undoubtedly inspired countless climbers throughout the Front Range and beyond.

At about the same time, famed 'Gunks hardman Richard Goldstone met Gill during a summer trip out west and was enormously impressed with Gill's one-arm pull-ups, front levers, and stiff boulder problems. Goldstone went back to the University of Chicago with an enthusiasm for training and adapted the use of surgical tubing (long utilized by gymnasts to build enough strength to do an Iron Cross) as a training aid for portions of his workouts. A few years later Goldstone returned east and became a significant presence during the rapid expansion of difficult free climbing at the Shawangunks. Goldstone discovered that Dick Williams (another former gymnast) was already training for climbing and incorporating dynamic movements in his campaign to free climb the many steep aid routes at the 'Gunks. Other Uberfall icons of the era such as Hans Kraus, Bonnie Prudden, Jim McCarthy, and John Stannard also had great interest and long personal histories in physical fitness. Kraus went on to form the President's Council on Physical Fitness, while Bonnie Prudden became a nationally recognized fitness expert and the first female athlete to appear on the cover of *Sports Illustrated*.

Meanwhile, it seemed California climbers were increasingly toying with one form or another of climbing training. Dave Rearick and Mike Sherick were specifically oriented toward gymnastics and could do presses into handstands and other gym stunts; Layton Kor was lifting weights regularly for his many outstanding ascents around Colorado and



German climbing icon Wolfgang Güllich demonstrating his one-arm power in Yosemite's Camp 4, circa 1980. GERD HEIDORN

in Yosemite. Remarkably, three of the top Yosemite free climbers of the day—Royal Robbins, Chuck Pratt, and Frank Sacherer—did very little training other than the usual regimen of pull-ups and push-ups. Jim Bridwell and Barry Bates followed in the spirit of their master-predecessors, training hard at such things as pull-ups on tree limbs in the Valley. Bates quickly developed the ability to do a one-finger pull-up with his middle finger on a sling hanging from a tree. The main thrust of their training, however, was simply to climb several days per week.

Bridwell, Bates, and others in the Camp 4 crowd were also likely influenced, directly or otherwise, by visiting climbers such as Goldstone and

Ament. Rich Goldstone is believed to have installed the first pull-up bar in Camp 4, while Ament brought the toughest bouldering discipline of the time to the Valley with his first ascents of 1968. Pat Ament also brought the slack chain to Yosemite, challenging climbers to develop refined balance and focus. The revered tradition of chain and rope walking in Yosemite began with the 40 feet of slack links that Ament strung between two Camp 4 trees. Bridwell and others went on to develop and deploy an array of training stations around Camp 4 so impressive that Warren Harding, the Yosemite Generation's sharpest wit, soon dubbed the area the "Olympic Training Village." Harding himself preferred to build stamina for his epic multiday

big-wall adventures by running to the top of Half Dome and back, a 17-mile round trip with nearly 1 mile of elevation change (and also, by his own admission, by refraining from hard liquor in the weeks before an ascent). In the years that followed, Camp 4 workout rigs introduced countless climbers from around the world to the basic elements of the future science of training for climbing.

Still, sports scientists in academia and the European mountain heartland had yet to view climbing as a subject worthy of serious and sustained study. Though ascents of the world's highest mountains were long a source of national pride in Europe, there were no Olympic medals (nor commercial sponsorships) to be won around which to build a culture of sport-specific training and achievement aimed at visible rewards. Climbing remained a rarefied pursuit, and research—where it pertained to climbing at all—was narrowly focused on the effects of long exposure to low-oxygen atmosphere. Still, the steadily growing popularity of climbing throughout the 1970s eventually gave birth to the first European studies relating to the physiological stresses and injuries associated with rock climbing.

In 1977 Pat Ament's *Master of Rock* was published. This biography of John Gill, though not focused specifically on training, served to document Gill's strength-training techniques and introduce them to a wider audience; the book quickly became not just an American classic but a kind of sacred writ for a new generation of climbers interested in pushing the absolute technical limits. *Master of Rock* opened a new door of consciousness, so to speak, of what it would take to be the best—not only in terms of physical ability but also in the broader context of performance—Gill being the epitome of performance excellence in virtually all his pursuits.

From the mid-1970s through the 1980s, the worldwide growth of technical rock climbing and the first climbing competitions produced an unprecedented exchange of ideas and innovations among European, Russian and Caucasian, and American



One of the first climbers to train with weighted pull-ups, John Bachar could pull up with nearly 140 pounds hanging from his waist! Here he trains with a "light" 50 pounds circa 1985.

PHIL BARD

climbers. In Yosemite's Camp 4; Boulder, Colorado; and the Shawangunks of New York, small groups trained and free climbed with increasing fervor, as energetic newcomers such as John Bachar, Jim Collins, Christian Griffith, Lynn Hill, Jim Holloway, John Long, Ron Kauk, Todd Skinner, Alan Watts, Tony Yaniro, and others arrived on the scene. Similarly, small groups of energetic climbers began to train in England, France, Italy, and Germany. The boulders of Fontainebleau and the ubiquitous limestone crags of the region became the proving grounds



The author (circa 1986) on his version of the “death board,” a training tool used by a handful of climbers in the pre-climbing-gym era.

HÖRST COLLECTION

for the first “sport climbers” in the early 1980s. The hard-training European climbers of the early sport-climbing era were Brits Ron Fawcett, Jerry Moffat, and Ben Moon; French icons Jibé Tribout, Antoine LeMénéstral, and Patrick Edlinger; Italians Roberto Bossi and Heinz Mariacher; and the powerful Germans Kurt Albert and Wolfgang Güllich.

In the United States no technique or aesthetic had a bigger impact on the rapid development of extreme free climbing than the import of sport-climbing tactics from Europe. Rappel-bolted routes eliminated the psychological stress and risks

associated with marginal protection, and through liberal use of hangdogging, the practitioner could safely work extreme sequences and thus bring Gill-level difficulty (5.13 moves) to roped climbing. At about this time, articles on physical performance and training began to appear in American climbing magazines; academic studies, too, began to proliferate, although initially focused on the subject of injuries specific to rock climbers. Strength-training techniques remained relatively unsophisticated, although a few key innovations such as the Bachar Ladder and fingerboard jacked generic finger and pull-power training up to a higher level of intensity and specificity.

In Europe’s sport-climbing culture, indoor walls had already taken hold, but it was not until 1987 that the first commercial climbing gym opened in the United States. Around the same time at the Campus Center—a weight-lifting facility at the University of Nürnberg—a strong German climber named Wolfgang Güllich developed a sport-specific form of reactive training known today simply as “campus training.” Between 1985 and 1991 Güllich went on to establish the world’s hardest free climbs and wrote a breakthrough training book, *Sportklettern Heute* (1986), and campus training quickly became a staple of elite climbers around the world. Toward the end of the century, as at its beginning, German climbers led the way to new levels of technical difficulty and athletic achievement.

The 1990s saw climbing go mainstream with televised competitions and dozens of well-sponsored full-time climbers in training year-round. The first two books on training for climbing by American authors were published in 1993 and 1994—Dale Goddard and Udo Neumann’s *Performance Rock Climbing* and *Flash Training* by this author—and articles on training became regular features of *Climbing* and *Rock and Ice* magazines. But the proliferation of indoor walls was the real wild card that allowed the average climber to practice more frequently and climb harder than ever before. All

the above-mentioned factors, along with improved equipment, made what was once the maximum grade, 5.10, quickly achievable by the masses; and remarkably 5.14 has become attainable by a handful of youngsters not even old enough to drive!

Beginning the new millennium, climbing is as popular as ever, as evidenced by outdoor crags now crowded with “weekend warriors” and the increasing number of semi-professional climbers training for World Cup competitions. A rapidly expanding knowledge base—professional climbing coaches, peer-review climbing research, and myriad online and print media on instruction and training—is now guiding recreational climbers into the lofty grades, once the domain of only a small group of professionals.

Top coaches such as Patxi Usobiaga, Patrick Matros, Dicki Korb, and Robyn Erbesfeld, among many others, are now leveraging new training technologies and the latest climbing research (as revealed throughout this book) to train current rock stars like Adam Ondra and Alex Megos. As a result these leading-edge climbers, along with veterans like Chris Sharma, Tommy Caldwell, and Daniel Woods, have recently advanced the cutting edge of climbing to V16 boulders, 5.15c/9b+ sport climbs, and, incredibly, 5.14d/9a Grade VII on the granite big walls of Yosemite.

As detailed in the previous pages, the rich century-plus history of training for climbing has delivered us to the exciting present era of evidence-based, technology-guided exercise programming. So while past generations trained largely in accordance with myth, anecdote, and trial and error, those entering the sport today have more credible information and more effective tools to leverage in training for the vertical extreme. (*Training for Climbing* is just one of several good texts now available, and I encourage you to explore some of the books on my Suggested Reading list at the end of this book.) Many of today’s up-and-coming climbers will quickly succeed at the once lofty grades



No one has climbed more 5.15a and harder routes than Adam Ondra, shown here training at home. CLAUDIA ZEIGLER

of 5.12/7b and 5.13/8a, while a subset possessing uncommon discipline and engaging in well-designed, long-term training programs have a good chance of reaching the 5.14a/8b+ benchmark . . . and for the genetically blessed, perhaps even 5.15d!

Genetics and Climbing Performance

Excuses are like parents—everybody has them. Ironically, your parents, or more precisely the genetic material you inherited from them, might be the best excuse why you or I may never climb 5.15. Still,

your genetic makeup, which substantially determines your height, flexibility, and natural strength, among other things, is a poor excuse for not being able to climb 5.10 or even 5.12. Yes, some specific 5.12 climb might require a long reach or high step that you will never be capable of making, but numerous research studies confirm my belief that the mass of climbers have the potential to succeed at the lofty grade of 5.12, regardless of genetics.

The Role of Genetics in Sports Performance

All other things being equal, genes seem to determine the differences in performance among individuals. In a sport as complex as climbing, however, you could argue that “all other things are never equal”—making the role of genetics in climbing performance hard to pin down. But let’s try!

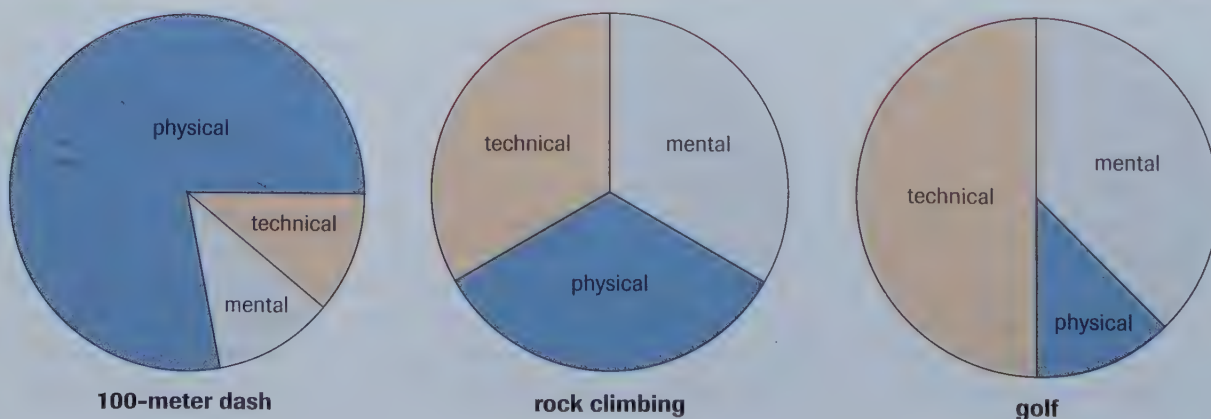
The role of genetics in what we become has been a favorite subject of scientists over the years—it’s the old “nature-versus-nurture” debate. Certainly genetics would seem to play an underlying role in our natural mental climate and personality. However, it’s in the physical realm of strength and motor skill that genetics appear to play the largest role (or at least this is where genetics seem most observable and measurable for scientists). Interestingly, a review of

the research on the role of genetics on performance reveals an extremely complex subject with contradictory theories and findings among academics. One study (Ericsson 1993) suggested that hours of deliberate practice are the most important factor in determining performance, while another (Fox 1996) found that genes are responsible for half the variations in performance among individuals.

In the real, nonacademic world, it seems that neither of these studies is unequivocally correct. It appears that genetics play the greatest role in sports where the raw physical demands far outweigh the mental or technical requirements. For example, excelling at the 100-meter dash requires extreme explosive power but only basic mental and technical skill. Conversely, golf requires mastery of a wide range of technical skills and a well-cultivated mental calm, but the physical demands are much less noteworthy. Therefore, while genetics clearly play a major role in determining who makes it to the Olympics in the 100-meter dash, they should have much less influence in determining who plays in this year’s PGA Championship.

Summing up: Hours of deliberate practice are a requisite for performing at a high level in complex (technical and mental) sports, whereas ideal genetics are a prerequisite for achieving greatness

Figure 1.1 Relative Demands of Various Sports



in the most physical pursuits, such as running and weight lifting. Rock climbing is unique among sports, however, in that it requires a near-equal balance of mental, technical, and physical prowess (see figure 1.1). So you can argue that genetics do play a significant, though not primary, role in determining your level of performance in this sport.

Genetic Factors Relating to Climbing Performance

So just what genetic factors might be helping or hurting you? I bet they are different and more subtle than you think. Height and weight seem to be what most climbers consider their blessing or curse, but it's likely a number of less obvious attributes that help make possible the incredible achievements of climber icons like Chris Sharma, Adam Ondra, and the Nicole, Huber, and LeMénéstral brothers.

While the aforementioned climbers exhibit a variety of body shapes and sizes, they all possess unusually high maximum grip strength, upper-body power, and/or local (forearm) endurance—beyond that which can be acquired by the average climber training “perfectly” for many years. The genetic gifts enabling these feats probably relate to hard-to-observe factors such as tendon insertions (where they originate from and insert into the bones of the hands and arms), lever length (length of bones), muscle fiber type, and hormone profiles.

With regard to tendon insertions, a slight shift in the location compared with normal provides additional leverage that gives a few lucky folks more grip strength (off the couch!) than others could achieve through years of training. Similarly, your innate ratio of fast-twitch to slow-twitch muscle fiber determines whether your natural aptitude tends toward high endurance, high strength, or neither. Finally, we each have unique hormone profiles (testosterone, cortisol, and so forth) that vary with age and sex, and this plays an underlying role in our response to training and ability to recover (Bloomfield 1994). Because of this, some people

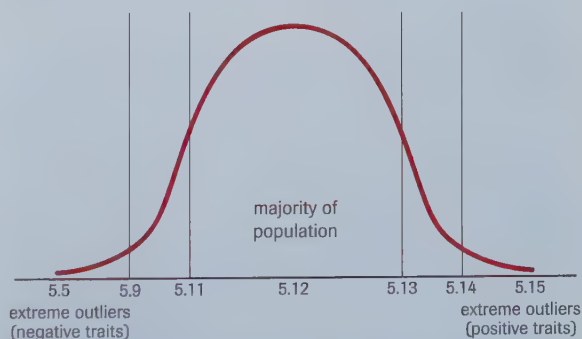
can climb hard three days in a row or respond more dramatically to training, while most of us need far more rest in order to perform well, and our training adaptations are more gradual.

If you still aren't convinced that genetics play a role in determining who will be the very best climbers, consider the three pairs of brothers mentioned earlier. Frederic and François Nicole, Alex and Thomas Huber, and Marc and Antoine LeMénéstral have all climbed at the fringe of maximum difficulty. This is not coincidence, but instead a screaming message that genetic makeup is a factor in climbing performance.

Your Genetic Potential as a Climber

If you're beginning to sense that you might lack some or all of the above genetic gifts, don't be depressed! As I stated earlier, odds are that you're “normal enough” to climb 5.12 and, if you climb a lot and train just right, deep into the 5.13 grade as well. Because of the large role that mental and technical skill plays in climbing performance, you can push very high up the grade scale by maximizing your capabilities in these areas. The bell curve (see figure 1.2) shows that most of the population falls

Figure 1.2 Genetic Potential



Adam Ondra and Daniel Woods would fall into the far right portion of the curve, whereas actors John Goodman and Verne Troyer would be far left.

in the middle of the bell, in the area representing near-average genetic characteristics. A much smaller number of folks—call them outliers (say, one in ten)—have somewhat better or worse genetics than average. Then there are the extreme outliers (say, 1 in 1,000 or more) who have the potential to be brilliant if they discover their gift and apply themselves completely.

The fact that most of us fall somewhere in the middle of the bell curve can be uplifting or depressing, depending on your perspective. If you dream of climbing 5.15 or V15, the chart shows that even if you do everything right and dedicate your entire life to it, the odds are low that you have the genetic potential to make this dream a reality. But if you currently climb 5.10 or V3 (or whatever), you should be psyched that 5.12—and likely even 5.13—is within your reach!

A few recent studies support this idea. One study (Barss 1997) divided a group of twenty-four recreational climbers into two groups based on climbing ability. With the exception of a straight-armed hang endurance test, there was no statistical difference in the performance of a wide variety of general and sport-specific tests between the “less skilled” group (those climbing 5.7 to 5.10a) and the “more skilled” group (those climbing 5.10b to 5.11b). Therefore, at the intermediate levels (5.7 to 5.11b), there’s a poor correlation between fitness and climbing ability. The stronger climbers were not necessarily the better climbers, so mental and technical differences account for the difference in ability.

Another, more complex study (Mermier 2000) looked at a larger group of forty-four male and female climbers with a wider range of abilities (5.6 to 5.13c). The results showed that the variance in climbing performance relates primarily to trainable variables, and that anthropometric variables (height, weight, arm and leg length, arm span, percent of body fat, and the like) are not a statistically significant factor. So this study also supports my sense that by optimizing technical and mental skills (the

trainable variables), the average climber should be able to progress to a high level of climbing, possibly even as high as 5.13c. Note that no 5.14 climbers were included in the study, so we don’t know if inclusion of these world-class individuals would have yielded similar results (I suspect not).

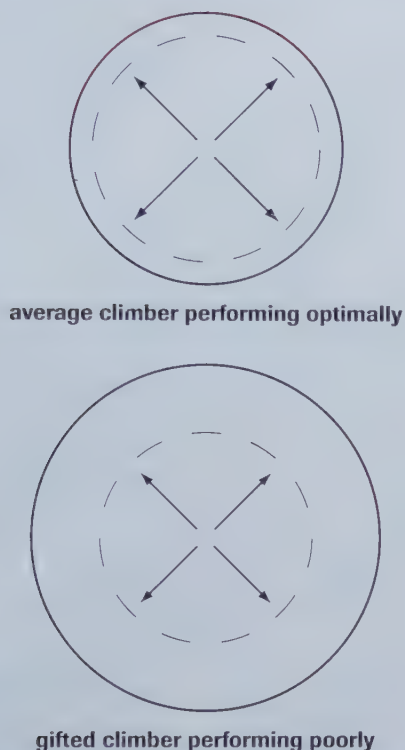
In fact, a third, very similar study (Watts 1993) looked only at world-class climbers (those competing in the semifinals at a World Cup event). It found that these elite individuals exhibited a higher grip-strength-to-body-mass ratio, had a lower percentage of body fat, and were of a slightly smaller stature when compared with other athletic groups. This study supports the idea that those world-class 5.14+/V15 climbers are born, not made, in that they are extreme outliers with just the right build to be able to climb at the highest levels of difficulty.

Great Genetics Don't Guarantee You'll Be a Great Climber

Still, the premise that climbing requires equal mastery of mental, technical, and physical abilities means that good genes aren’t enough to make you a rock star. Just as genetically average individuals can progress to climbing 5.12 or 5.13 by perfecting their technical and mental skill sets, genetic freaks who can crush bricks in their hands may forever remain “average” climbers due to poor technique or lack of mental skills.

Consider figure 1.3, which depicts the genetic potential (solid line) and real-life ability (dashed line) of a climber with average genetic makeup versus the brick-crushing genetic freak. Through dedicated, intelligent training of all the elements under her control, the average climber has pushed her ability almost the whole way out to her genetic potential. The superstrong genetic freak, on the other hand, with his poor technique and mental control, is an underachiever when you compare his real-life performance with his genetic potential. Comparatively, the genetically average climber pushed the dashed line out farther and is thus the real master of rock!

Figure 1.3 Genetic Potential



Genetic potential (solid line) versus real-life ability (dashed line). Who's the peak performer?

Next time you go to the gym or crag, observe all the men and women, of all ages, shapes, and sizes, who are climbing 5.11, 5.12, and even 5.13. The vast majority of these folks are of average genetic makeup (located near the middle of the bell curve in figure 1.2), but through dedication and hard work on all aspects of the climbing game, they have succeeded at pushing their dashed line out toward the edge of their genetic limitations.

Limits to Climbing Performance

The top climbing grade exploded upward from 5.12d in the mid-1970s to 5.14b by the end of the 1980s. The primary reasons for this marked

improvement are equipment (better shoes, stickier shoe rubber, easy-to-place active camming devices, and bolt-protected routes), better training (indoor walls, fingerboards, and so on), and more effective practice methods (hangdogging). In 1991 Wolfgang Güllich upped the ante with his ascent of Action Directe (5.14d/9a) in Germany. Incredibly, this route went unrepeated for five years and has since seen only about one ascent per year, despite being the world's most famous sport climb.

Now, more than twenty-five years after the first ascent of Action Directe, the benchmark of maximum difficulty has been stretched to 5.15c and V16—but the advancement has been relatively



Wolfgang Güllich on his breakthrough route Action Directe, the world's first 5.14d.

THOMAS BALLEMBERGER



*Melissa Le Nave training
in the one-of-a-kind system
wall at the Bloc Out gym
in Fontainebleau, France.*
CLAUDIA CRIVIERI

slow, especially in bouldering, which most directly tracks the absolute maximum climbing level. After Güllich's first ascent of Action Directe, it took a full decade until the first consensus 5.15a/9b, Realization, was established by Chris Sharma in 2001. Seven more years passed before Sharma completed his long-term project, Jumbo Love, widely considered the world's first 5.15b/9b+. However, only a comparatively short four years was needed for Adam Ondra to establish 5.15c/9c with his mega roof route, Change, at Flatanger, Norway.

Given the length of time required to consolidate the 5.15 grade—and even more remarkable, the slow push into the realm of V16 by top boulderers—we have to wonder if we are approaching human limitations to free climbing. While we can never rule out another breakthrough in technology (equipment), it's highly unlikely that we will see another grade explosion as occurred during the 1980s. Instead, slow increases will occur over the time frame of decades.

For a glimpse of what we might expect, let's look at several “mature” Olympic events. Over the last half century, improvements per decade have been approximately as follows: sprinting—1 percent, distance running—1.5 percent, jumping—3 percent, pole vaulting—5 percent, swimming—5 percent, skiing—10 percent (Seiler 2000). Improved equipment surely contributed to the higher values for pole vaulting (fiberglass poles), swimming (“frictionless” speed suits), and skiing (ski technology seems to be constantly improving). Unfortunately, performance-enhancing drugs are also a very real factor in the improvements in many Olympic events.

Assuming no technological breakthroughs and no drugs (though I'm sure there are some), a good bet would be that the top climbing level would increase by just a few percent per decade. I believe these gains will result from the identification of more extreme outliers as participation in climbing increases, and from better matching of appropriate

training on a more individual basis. The recent breakthrough ascents of a small crop of Generation Z superstar climbers—Adam Ondra, Alex Megos, Ashima Shiraishi—are a result of the intersection of talent identification, high-quality coaching, and a near single-minded focus. Ondra has now established three 5.15c/9c routes; Alex Megos, at age nineteen, achieved the world's first 5.14d/9a on-sight and soon thereafter dispatched Action Directe and Biography in just a few tries; and at the age of thirteen, Ashima Shiraishi achieved her first V14 boulder and completed the first 5.15a redpoint ascent by a woman!

While 5.15d may soon become a reality, it could be that our current method of rating routes is on the verge of breaking down—the difference between 5.15b and 5.15c (or V15 and V16) may be purely a function of anatomical variation in the context of a single move or sequence. This argues for an entirely different system of assessing difficulty at the top levels, such as a scale that counts the number of climbers able to do a given move or problem. Interestingly, this is the essence of John Gill's B-scale for grading bouldering problems, developed back in the late 1950s.

Training for Climbing

As discussed earlier, there are many trainable variables to work on as part of your training for climbing program. In chapter 2 you will perform a self-assessment test to determine which of these trainable variables is most holding you back. The best training program (for you) will concentrate on the areas that can produce the greatest gain in performance output for a given training input. Of course, the goal is to train most effectively, not maximally.

A Definition of Training for Climbing

I define training for climbing as any practice, exercise, or discipline that increases absolute climbing performance. Clearly, this represents a broad

spectrum of subjects—hence the wide range of topics covered in this book.

Through this paradigm you should recognize that training includes a wide range of activities and practices such as bouldering (to learn problem solving and develop power); climbing on a home wall or at a climbing gym (to improve technique and strength); on-sighting, hangdogging, or for that matter any climbing (to enhance your mental and physical skill sets); and traveling to experience many different types of climbing (to gain experience and a broad range of technical skills). Training also includes efforts made in ancillary areas such as stretching and antagonist-muscle training (for flexibility and injury prevention), eating properly (to enhance recovery and maintain optimal body composition), visualization and targeted thinking (to maximize mind programming and disconnect from bad habits), resting sufficiently and listening to your body (to optimize training results and avoid injury), and evaluating yourself regularly (to determine your current strengths and weaknesses). Finally, of course, training includes proper execution of various general and climbing-specific exercises (to work toward your physical genetic potential).

For the sake of discriminating among these many types of training throughout the rest of the book, let's define several training subtypes—mental training, skill practice, fitness and strength training, and training support activities—as shown in figure 1.4.

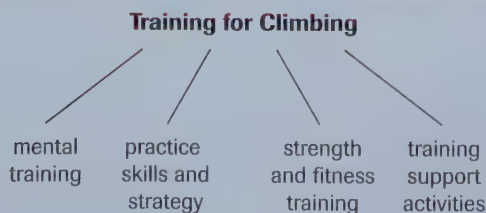
Mental training involves any thought control, discipline, or mind-programming activity that will directly or indirectly impact your climbing in a

positive way. The best climbers train mentally every day—this is one activity in which you can never overtrain—by targeting their thoughts only on things that can, in some way, influence their climbing and by deleting thoughts and habits that might hold them back. Unfortunately, many individuals possess mental muscle that's in an advanced stage of atrophy from underuse. Visualization is just one of the many mental exercises that can improve your climbing. Chapter 3 lays out an array of mental-training methods and on-the-rock strategies that will have a combined effect similar to unloading a heavy weight from your back (which you've unknowingly been hauling up routes). Are you ready to spread your Mental Wings?

Skill practice relates to time spent learning and refining actual sport skills and strategies outside of a performance setting. Just as baseball, basketball, and football players spend many hours practicing their skills outside of competition, climbers must practice by climbing a lot with the sole intention of improving climbing skill (and not worrying about an outcome such as a flash, redpoint, or on-sight ascent). It's my sense that many climbers' training programs are devoid of this vital subtype of training. We'll take an in-depth look at the subject of effective skill practice in chapter 4.

Fitness and strength training covers a wide range of activities that are performed with the primary intent of improving physiological capabilities. This includes general conditioning activities such as running, stretching, and light free-weight training as well as the more important climbing-specific activities such as fingerboard, campus, and hypergravity training. Many other activities can fall under this heading, as long as they somehow help improve your climbing performance or prevent injury. It's surprising, however, how many things done in the name of training for climbing actually have a negative effect on climbing performance. Get ready to sort things out as we take a cutting-edge look at physical training for climbing in chapters 5 through 9. Then, in chapter

Figure 1.4 Subtypes of Training



10 you will be guided on developing an effective and time-efficient personalized training program.

Finally, training support activities comprise a variety of crucial, yet often overlooked (or ignored), issues outside of your actual physical practice and training for climbing. Athletes in many other sports have known the vital role that rest, nutrition, and recovery acceleration techniques play in their ultimate level of performance. Serious climbers looking to press out their ability level toward their genetic limit must act on these issues with utmost discipline. Chapters 11 and 12 cover these important topics—applying the material may be the key to succeeding on your own “personal Action Directe”!

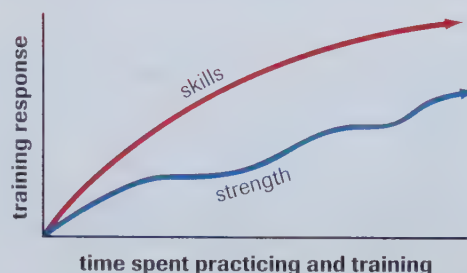
The Relationship Between Skill and Fitness

While the various subtypes of training for climbing will be discussed separately, they clearly affect one another. This is especially true when it comes to skill practice and fitness and strength training, so let's dig a little deeper.

For a beginning climber in the earliest stages of learning, a low level of fitness can slow the learning of climbing skill. A certain level of strength is necessary in order to practice enough (i.e., climb) to develop the basic skills of movement, hand- and footwork, and body positioning. Conversely, too much strength enables a beginner to get by on easy to moderate routes despite inefficient movement, poor footwork, and improper body position. Obviously, this will also slow (or prevent) the development of good technique—unless, that is, the strong person makes good technique the primary goal, instead of just getting up the route no matter what.

The problem is further aggravated by the fact that people tend to develop their talents disproportionately. Strong people are most likely into strength training, flexible people probably stretch regularly, and skillful people undoubtedly climb a lot. Sure, the drudgery of working on weak points isn't fun, and at times it can be discouraging. But if you really

Figure 1.5 Relative Gains



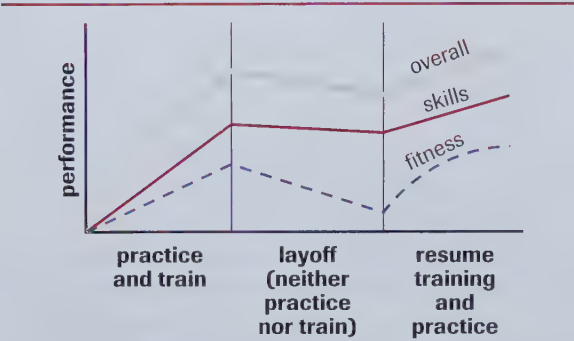
Relative gains from skill practice and strength training.

want to climb harder, you must train smarter. That means knowing where to best invest your time to get the most output for your training input. For the majority of climbers, the best investment is on further development of climbing skills and strategy (see figure 1.5).

Elite climbers may have less to gain from practicing familiar forms of climbing. These expert climbers are way out on the learning curve near their ultimate skill potential, so maximum strength (and the mind) become the crucial factor in performance. Hence, we commonly see magazine articles about these rock stars that describe seemingly lethal or disastrously stressful strength-training regimes that would surely send the ordinary climber into a state of overtraining, the doctor's office, or a self-defeating over-reliance on strength training as the key to improvement.

Focused strength training is of greater importance for all climbers after a layoff, whether due to injury, winter, or some other reason (see figure 1.6). The rapid loss of fitness that occurs when training or climbing ceases for a period of weeks or months is best counteracted by several weeks of dedicated physical training (fortunately, you largely maintain climbing skill once the motor programs are well established). While this short-term training focus helps in regaining your old form, the long-term and

Figure 1.6 Performance Losses



Performance losses (gains) during (and after) a layoff.

most significant improvements in climbing ability will still result from effective skill practice until late in your career. Only at the lofty grades of 5.12 and above does climbing-specific strength become a major limiting factor.

Specific Adaptation to Imposed Demands (SAID)

Serious climbers would be wise to train and climb in accordance with the cornerstone principles of the field of exercise science. For example, knowledge of the SAID Principle (specific adaptation to imposed demands) can be leveraged to maximize the effectiveness of your training for a specific climbing goal or dream climb.

The SAID Principle explains that a certain exercise or type of training produces adaptations specific to the activity performed and only in the muscles (and energy systems) that are stressed by the activity. For instance, running produces favorable adaptations in the leg muscles and the cardiovascular system. But the muscles and systems not stressed

show no adaptation, so even heroic amounts of running will produce no favorable changes in, say, the arms. Of course, the adaptations that result from running do transfer somewhat to other activities that depend on the same body parts and systems (such as mountain biking or hiking). **Bottom line:** The SAID Principle demands that effective training for climbing must target your body in ways very similar to climbing (body position, muscles used, energy systems trained, and so forth).

Similarly, your body adapts in a specific fashion to the specific demands you place on it while climbing. If you boulder a lot, you will adapt to the specific skill and strength demands of bouldering. If you climb mostly one-pitch sport routes, you adapt to the unique demands of zipping up, say, 30 meters of rock before muscular failure. If you favor multi-pitch routes or big walls, your body will adapt in accordance to the demands of these longer climbs. Or, if your outings are alpine in nature, your physiological response will be specific to the very unique demands of climbing in the mountains.

The vitally important distinction here is that while all these activities fall under the headline of “climbing,” they each have unique demands that produce very specific physical adaptations. Therefore, the training effect from regular bouldering will do very little to enhance your physical ability for alpine climbing. Figure 1.7 shows that the specific demands of sport climbing are much closer to those of bouldering. Consequently, the adaptations incurred from frequent bouldering will largely carry over to sport climbing (especially short sport climbs) and vice versa.

Due to the SAID Principle, your practice and training on the rocks should be spent mostly on the type of climbing in which you wish to excel. It’s no

Figure 1.7 Continuum of Specific Adaptations for Various Subdisciplines of Climbing



mistake that the best boulderers in the world rarely tie in to a rope. Likewise, the best alpine climbers spend little time working thirty-meter sport routes. Targeting your training on the specific demands of your preferred form of climbing is the essence of the SAID Principle.

In the end you must make a philosophical choice whether you want to specialize—and therefore excel—in one or two of the climbing subdisciplines, or become a moderately successful all-around climber. Certainly there is merit and reward in both approaches.

Summary of Training for Climbing

1. Training for climbing is any practice, exercise, or discipline that increases absolute climbing performance.
2. Mental training begins by increasing your commitment to all things climbing (while reducing time, energy, and thought invested in lower-value activities and hobbies) and continues with the development of uncommon self-awareness and superior thought control and risk management skills.
3. Skill practice is paramount, since climbing skills and tactics are distinctly unique from those of other sports. Only going climbing will make you a more skilled climber.
4. Fitness training comprises both general and sport-specific exercises and activities. Novice climbers should engage in a period of general conditioning before advancing to the more stressful climbing-specific exercises.
5. Training support activities are essential for all serious climbers, including scheduled rest days, proper nutrition, and use of techniques to accelerate recovery.
6. Ultimately, the most effective training-for-climbing program will target limiting constraints in highly climbing-specific ways (in accordance with the SAID Principle).

A full-page photograph of a woman, Paige Claassen, climbing a massive, reddish-brown rock face. She is positioned in the upper-middle section of the frame, wearing a blue tank top and dark pants. The rock face is steep and textured with horizontal and vertical fissures. Below the climber, a small town with white buildings and green trees is visible, along with a large, open field. The sky is clear and blue.

***Paige Claassen climbing
Ganesh (5.14a) Badami, India.***

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